

Solutions to Homework #6

Problem 1

- (a) The anti-symmetry assumption implies $h[n] = -h[L-1-n]$. In particular, we have $h[(L-1)/2] = 0$. Then the frequency response is given by

$$\begin{aligned}
 H(e^{j\omega}) &= \sum_{n=0}^{L-1} h[n] e^{-j\omega n} \\
 &= \left(\sum_{n=0}^{(L-1)/2-1} h[n] e^{-j\omega n} \right) + \left(\sum_{m=(L-1)/2+1}^{L-1} h[m] e^{-j\omega m} \right) \\
 &= \left(\sum_{n=0}^{(L-1)/2-1} h[n] e^{-j\omega n} \right) + \left(\sum_{n=0}^{(L-1)/2-1} h[L-1-n] e^{-j\omega(L-1-n)} \right) \\
 &= \left(\sum_{n=0}^{(L-1)/2-1} h[n] e^{-j\omega n} \right) + \left(\sum_{n=0}^{(L-1)/2-1} (-h[n]) e^{-j\omega(L-1-n)} \right) \\
 &= \sum_{n=0}^{(L-1)/2-1} h[n] \left(e^{-j\omega n} - e^{-j\omega(L-1-n)} \right) \\
 &= \sum_{n=0}^{(L-1)/2-1} h[n] e^{-j\omega \frac{(L-1)}{2}} \left(e^{j\omega(\frac{L-1}{2}-n)} - e^{-j\omega(\frac{L-1}{2}-n)} \right) \\
 &= \sum_{n=0}^{(L-1)/2-1} h[n] e^{-j\omega \frac{(L-1)}{2}} 2j \sin \left(\left(\frac{L-1}{2} - n \right) \omega \right) \\
 &= \sum_{m=1}^{(L-1)/2} h \left[\frac{L-1}{2} - m \right] e^{\frac{-j\omega(L-1)}{2}} 2j \sin(m\omega) \\
 &= \sum_{m=1}^{(L-1)/2} h \left[\frac{L-1}{2} - m \right] e^{\frac{-j\omega(L-1)}{2}} 2e^{-j\frac{\pi}{2}} \sin(-m\omega) \\
 &= e^{-j(\frac{\omega(L-1)}{2} + \frac{\pi}{2})} \sum_{m=1}^{(L-1)/2} h \left[\frac{L-1}{2} - m \right] 2 \sin(-m\omega).
 \end{aligned}$$

Therefore,

$$A(\omega) = \sum_{m=1}^{(L-1)/2} h \left[\frac{L-1}{2} - m \right] 2 \sin(-m\omega)$$

and

$$\phi(\omega) = \frac{\omega(L-1)}{2} + \frac{\pi}{2}.$$

Since $\ell = \frac{L-1}{2}$ satisfies $2\ell \in \mathbb{Z}$ and $\phi_0 = \frac{\pi}{2}$, we deduce that $h[n]$ has generalized linear phase.

(b) The DC gain is computed as

$$H(e^{j0}) = \sum_{n=0}^{L-1} h[n] = h[(L-1)/2] + \sum_{n=0}^{(L-1)/2-1} (h[n] + h[L-1-n]) = 0.$$

Here we split the sum into the center and the sub-summations of the other indices. All summands become zero due to the anti-symmetry assumption.

(c) The high frequency gain is computed as

$$\begin{aligned} H(e^{j\pi}) &= \sum_{n=0}^{L-1} (-1)^n h[n] \\ &= (-1)^{(L-1)/2} h[(L-1)/2] + \sum_{n=0}^{(L-1)/2-1} ((-1)^n h[n] + (-1)^{L-1-n} h[L-1-n]) \\ &= (-1)^{(L-1)/2} h[(L-1)/2] + \sum_{n=0}^{(L-1)/2-1} ((-1)^n h[n] + (-1)^n h[L-1-n]) \\ &= (-1)^{(L-1)/2} h[(L-1)/2] + \sum_{n=0}^{(L-1)/2-1} (-1)^n (h[n] + h[L-1-n]) = 0. \end{aligned}$$

Here we used the fact that $L-1$ is an even number, which implies $(-1)^{L-1} = 1$.

Problem 2

(a) Recall that the bilinear transform is given by

$$s = \frac{1 + z^{-1}}{1 - z^{-1}}.$$

Let s be any complex number on the $j\Omega$ axis, that is, the real part of s is zero or s is a pure imaginary number. Let us rewrite s as $s = j\Omega$ for $\Omega \in \mathbb{R}$. Then the image of s via the inverse bilinear transformation is given as

$$z = \frac{s+1}{s-1} = \frac{j\Omega+1}{j\Omega-1}$$

and it satisfies

$$|z| = \frac{|j\Omega+1|}{|j\Omega-1|} = \frac{\sqrt{1+\Omega^2}}{\sqrt{1+\Omega^2}} = 1.$$

- (b) Let s be a complex number such that $\text{Re}(s) < 0$, that is, located in the left-half s -plane. It suffices to show that every such s is mapped to the inside of the unit circle in the z -domain. Let $s = \sigma + j\Omega$ and assume that $\sigma < 0$. Then the image of s via the inverse bilinear transform satisfies

$$|z| = \frac{|j\Omega + \sigma + 1|}{|j\Omega + \sigma - 1|} = \frac{\sqrt{(1 + \sigma)^2 + \Omega^2}}{\sqrt{(1 - \sigma)^2 + \Omega^2}} < 1.$$

Here we used the fact that $(1 + \sigma)^2 < (1 - \sigma)^2$ due to $\sigma < 0$. Therefore, the left-half s -plane is mapped to the inside of the unit circle in the z -domain.

- (c) Note that $e^{j\omega}$ is mapped to $s = j\Omega$ by

$$j\Omega = \frac{1 + e^{-j\omega}}{1 - e^{-j\omega}} = \frac{e^{j\omega/2} + e^{-j\omega/2}}{e^{j\omega/2} - e^{-j\omega/2}} = \frac{\cos(\omega/2)}{j \sin(\omega/2)}.$$

Therefore,

$$\Omega = -\frac{1}{\tan(\omega/2)}.$$

Note that $\omega = \pi$ is mapped to $\Omega = 0$. As $\omega > 0$ decreases to 0, $\Omega < 0$ decreases to $-\infty$. Furthermore, as $\omega < 0$ increases to 0, $\Omega > 0$ increases to ∞ . This means that high discrete-time frequency corresponds to low continuous-time frequency. Therefore, the map converts a continuous-time lowpass filter to a discrete-time highpass filter. The cut-off frequencies are determined by

$$\Omega_s = |\tan(\pi/6)^{-1}| = \sqrt{3}, \quad \Omega_{p1} = |\tan(\pi/2)^{-1}| = 0, \quad \Omega_{p2} = |\tan(\pi/4)^{-1}| = 1.$$

Therefore, the specifications of the corresponding continuous lowpass filter is given by

$$0.95 \leq |H_c(j\Omega)| \leq 1.05, \quad 0 \leq |\Omega| \leq 1$$

and

$$|H_c(j\Omega)| \leq 0.01, \quad |\Omega| \geq \sqrt{3}.$$

Problem 3

We are given

$$H(z) = H_c(s) \Big|_{s=\beta \left[\frac{1-z^{-\alpha}}{1+z^{-\alpha}} \right]},$$

where α is a nonzero integer and β is a real number.

- (a) The statement holds true for every $\beta > 0$. Recall that

$$s = \beta \left[\frac{1 - z^{-\alpha}}{1 + z^{-\alpha}} \right].$$

By solving for z^α , we obtain

$$z^\alpha = \frac{\beta + s}{\beta - s}.$$

Let $s = x + jy$. Then

$$|z| = \left(\frac{(x + \beta)^2 + y^2}{(x - \beta)^2 + y^2} \right)^{\frac{1}{2\alpha}}$$

Since we assume that $\beta \neq 0$, it follows that $x = 0$ holds if and only if $|z| = 1$. In other words, the $j\Omega$ axis in the s -plane is mapped to the unit circle in the z -plane. Furthermore, since $\beta > 0$, we obtain that

$$x < 0 \iff |z| < 1$$

and

$$x > 0 \iff |z| > 1.$$

Therefore, the left-half plane in the s -plane is mapped to the interior of the unit circle in the z -plane.

Since the poles of a stable, causal, continuous-time filter are located in the left half space, they are mapped to the interior of the unit circle in the z -plane. Therefore, the statement holds true.

- (b) The statement holds true for every $\beta < 0$. Similar to part (a), we have

$$|z| = \left(\frac{(x - \beta)^2 + y^2}{(x + \beta)^2 + y^2} \right)^{-\frac{1}{2\alpha}}$$

The $j\Omega$ -axis is still mapped to the unit circle. Furthermore, since $\beta < 0$ and $-\frac{1}{2\alpha} > 0$, we have

$$x < 0 \iff |z| < 1$$

and

$$x > 0 \iff |z| > 1.$$

By the same argument, the statement holds true.

- (c) By evaluating the transform at $s = j\Omega$, we obtain

$$z^2 = \frac{1 + j\Omega}{1 - j\Omega}.$$

Therefore it follows that

$$|z| = \left(\frac{1 + \Omega^2}{1 + \Omega^2} \right)^{1/4} = 1.$$

In other words, the $j\Omega$ -axis in the s -plane is mapped to the unit circle in the z -plane.

(d) By restricting s and z respectively to $j\Omega$ and $e^{j\omega}$, we obtain the map

$$j\Omega = \frac{1 - e^{-j2\omega}}{1 + e^{-j2\omega}} = \frac{e^{j\omega} - e^{-j\omega}}{e^{j\omega} + e^{-j\omega}} = \frac{2j \sin(\omega)}{2 \cos(\omega)} = j \tan(\omega),$$

which implies

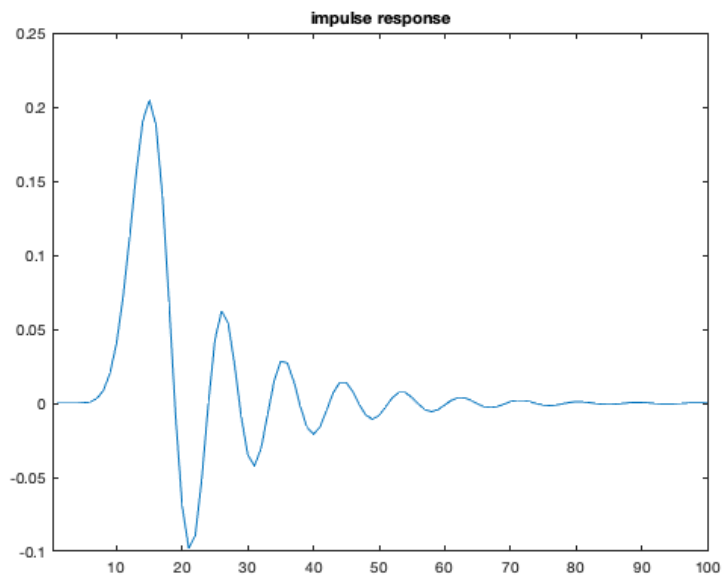
$$\Omega = \tan(\omega).$$

Since $\tan(\cdot)$ is π -periodic, the inverse image of $[-1, 1]$ is given by

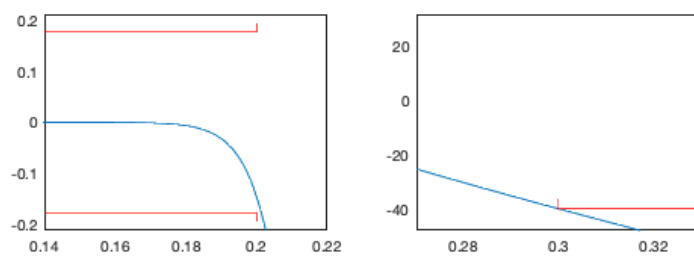
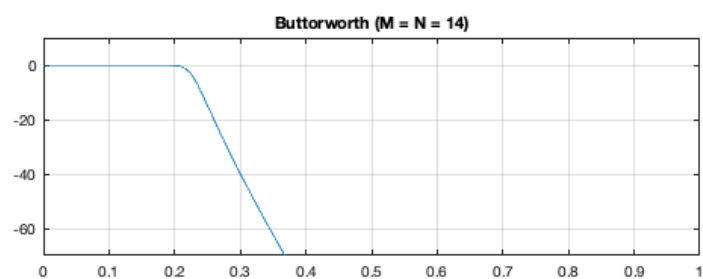
$$\tan^{-1}([-1, 1]) = \left\{ \omega : |\omega| \leq \frac{\pi}{4} \right\} \cup \left\{ \omega : \frac{3\pi}{4} \leq |\omega| \leq \pi \right\}.$$

Problem 4

(a) Butterworth filter

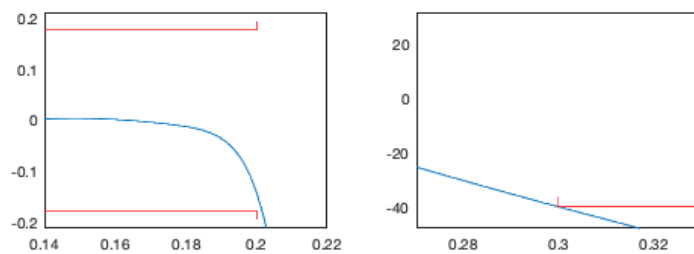
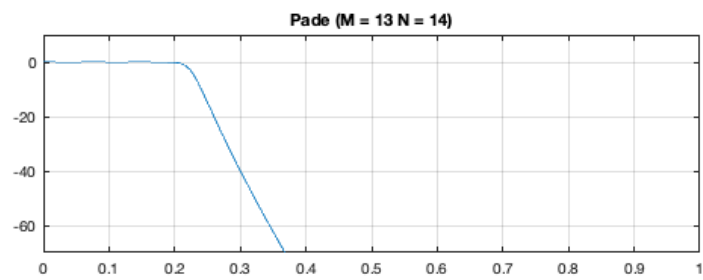


Impulse response



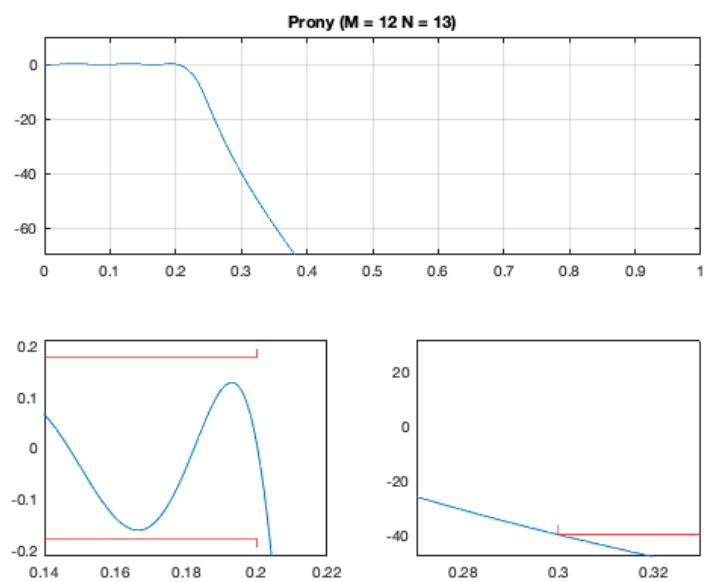
Frequency response

(b) Pade's approximation



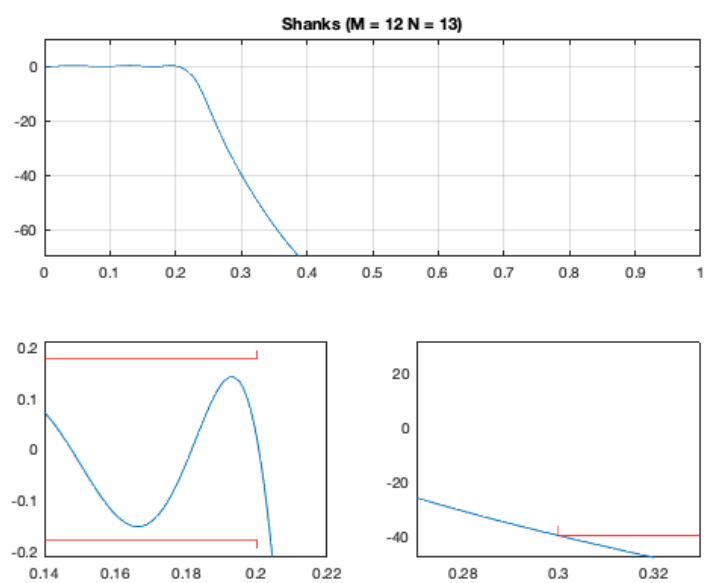
Frequency response

(c) Prony's method



Frequency response

(d) Shank's method



Frequency response with Shank's approximation